



Acoustics Lite - Documentation

Version: 1.0

Compatibility: Unity 2021.3+ (Tested on Unity 6000.0.58f2)

Quick Start window can be found under **Tool/Acoustics Lite/Help Window**

My website : <https://audran-demeyer-gramond.my.canva.site>

1. Overview

Acoustics Lite is a lightweight audio solution for Unity that provides realistic, real-time sound obstruction without the need for complex baking or pre-calculation. It uses smart raycasting to simulate how sound behaves when blocked by obstacles.

Key Features

- **Real-Time Occlusion:** Sounds are dynamically muffled by walls using a LowPass Filter and volume attenuation.
- **Volumetric Source Detection:** Unlike standard raycasts that come from a single point, this system simulates a "radius" for the sound source. A large sound source is harder to block completely than a small one.
- **Diffraction Simulation:** Simulates sound "wrapping" around corners and pillars.
- **Indirect Paths:** Detects if sound can bounce through a doorway or opening to reach the player, even if there is no direct line of sight.

2. Installation & Setup

Step 1: Scene Physics

Ensure your walls, floors, and obstacles have Colliders attached.

Assign these objects to a specific Layer (e.g., "Default" or a custom "Obstacles" layer). This is crucial for the raycasting system to detect them.

Step 2: Setting up Sound Sources

1. Select any GameObject that emits sound (e.g., Radio, Enemy, Footsteps).
2. Add the **AcousticSource** component.
3. *Note:* The script will automatically require and add a standard Unity `AudioSource` and `AudioLowPassFilter` if they are missing.
4. Assign your **Player** or **Main Camera** (wherever the `AudioListener` is located) to the `Listener` field in the inspector. (If left empty, it will try to auto-detect the Main Camera).
5. Set the **Occlusion Layer** to the layer you chose in Step 1 (e.g., "Default").

3. Component Reference

AcousticSource

This is the main component that handles all real-time calculations.

Configuration

- **Listener:** The target the sound is traveling to (usually the Player).
- **Occlusion Layer:** Which physical layers block sound.

Occlusion Settings

- **Lerp Speed:** (Default: 8.0) How quickly the sound transitions between muffled and clear. Higher values = faster transitions.
- **Min Volume Ratio:** (Default: 0.3) The volume percentage maintained when the sound is fully blocked by a wall.
 - **0.0:** Completely silent behind walls.
 - **0.3:** Muffled but audible (realistic simulation of bass passing through walls).

Volumetric Source (Direct Path)

- **Source Radius:** (Default: 1.0) The physical size of the sound emitter.
 - **Small (0.1):** Easily blocked by thin pillars or small objects. Good for small sounds like a phone ringing.
 - **Large (1.5):** Harder to block completely; the sound will "wrap" around corners more easily. Good for large machinery or explosions.
- **Direct Ray Count:** (Default: 8) The number of rays cast to check visibility. Higher = more accurate diffraction but higher CPU cost.

Indirect Simulation

- **Indirect Ray Count:** (Default: 16) The number of rays cast to find "holes" (doorways/windows) in the environment.
- **Scatter Radius:** (Default: 6.0) How far around the player the system looks for indirect paths.
- **Bounce Penalty:** (Default: 0.15) The volume reduction applied when sound takes an indirect path (reflects/bounces) to reach the player.

4. Troubleshooting / FAQ

Q: My sound is completely silent when I go behind a wall.

A: Check your Min Volume Ratio. If it is set to 0, the sound will mute completely upon occlusion. Try setting it to 0.3 or 0.4 to allow some sound to bleed through. Also, ensure your AudioSource Spatial Blend is set to 3D (slider to the far right).

Q: The sound cuts out instantly instead of fading smoothly.

A: Check your Lerp Speed. If it is set very high (e.g., 20+), the change happens instantly. Lower it to 5 or 8 for a smoother transition.

Q: The system is ignoring my walls.

A: Check the Occlusion Layer on the AcousticSource component. It must match the Layer assigned to your wall GameObjects. Also, ensure your walls have Colliders.

Q: Performance is low with many audio sources.

A: Reduce the Direct Ray Count (try 4) and Indirect Ray Count (try 8) on less important sound sources. You can also disable "Indirect Simulation" entirely for minor background sounds by setting Indirect Ray Count to 0.

5. Architecture & Logic (Advanced)

Diffraction Logic

Standard audio occlusion usually checks a single line from Point A to Point B. If blocked, the sound stops.

Acoustics Lite casts rays from a volume (Sphere) around the source. It calculates a "Visibility Factor" (0.0 to 1.0).

This factor is raised to a power ($\text{Mathf.Pow}(\text{vis}, 0.25)$), ensuring that even a small 10% line-of-sight results in ~56% volume. This simulates how low-frequency sound waves diffract (bend) around obstacles rather than being cut off linearly.

Indirect Paths

When the direct line of sight is blocked ($\text{Visibility} < 60\%$), the system fires secondary "Scatter Rays" toward the listener's surroundings. If these rays hit a wall and can then "see" the player, the system confirms a "Reflection Path" exists (like sound bouncing off a corridor wall through a door). This restores volume and high frequencies that would otherwise be lost.